

2-⑥

(II)

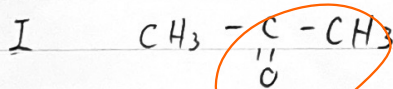
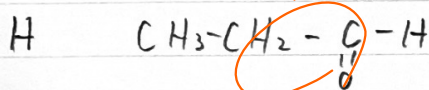
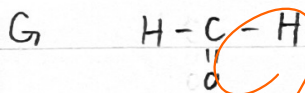
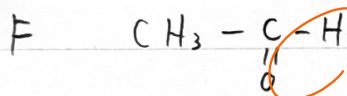
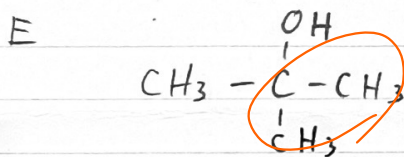
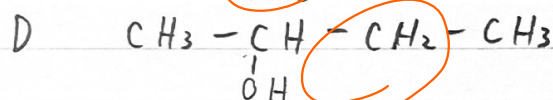
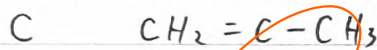
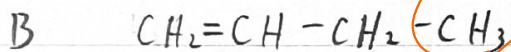
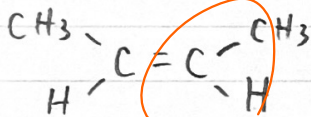
1、ア 酢酸化銅; 試料の不完全燃焼を防ぐ

イ 塩化カルシウム; 水を吸収する

ウ ソーダ石灰; 二酸化炭素を吸収する

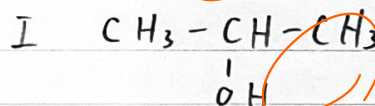
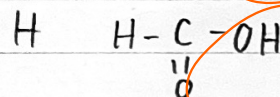
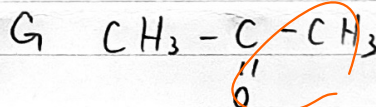
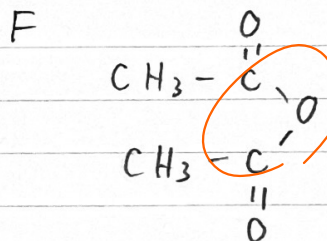
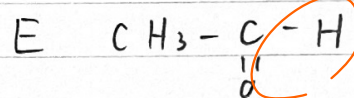
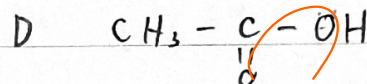
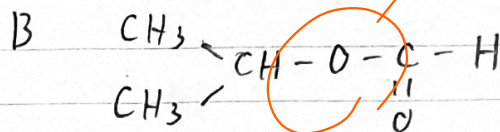
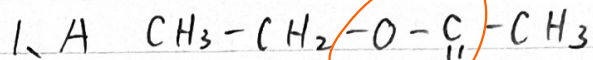
2、 C_4H_8

3、A



4、F

3-⑭



2、ア 水酸化ナトリウム 1 ヨウ素

ヨードホルム反応; CHI_3

3、銀鏡反応

4、アセトアルデヒド

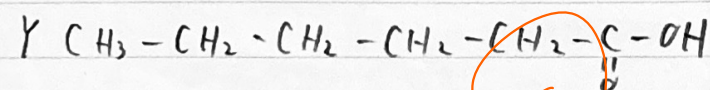
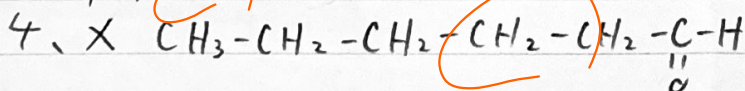
3-⑬

1、(a) 8 種類

(b) 6 種類

2、3 種類

3、4 種類



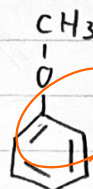
4-⑬

1、 C_7H_8O

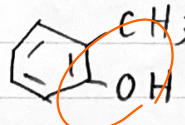
2、A



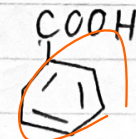
B



C

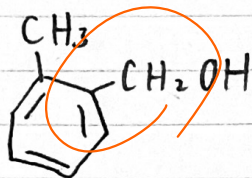


D

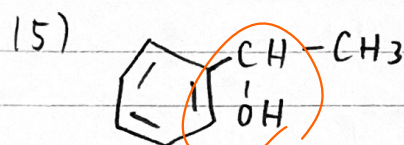
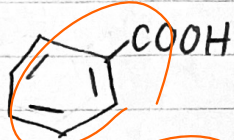


4-14

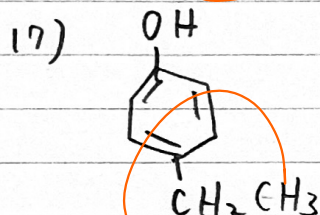
(1) 4 (2)

(3) CHI_3

(4)



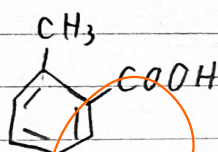
(16) 9



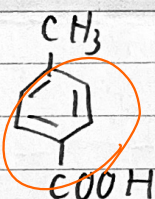
(18) 5 (19) 1:2:2:2:3

4-16

1. (1) A

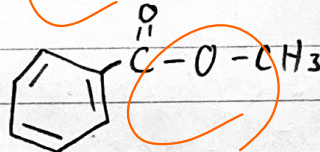


B

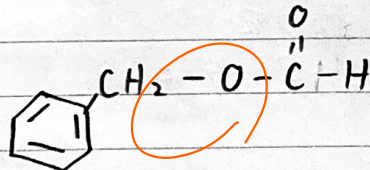


(2) o-キシレン、+フタレン

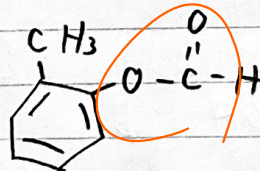
2. (1) C



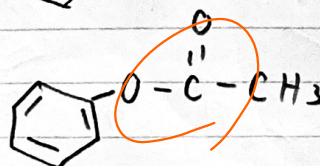
D



E



F

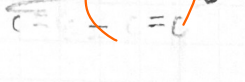
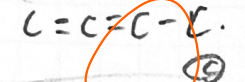
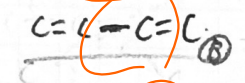
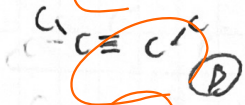
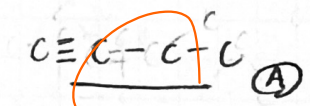


(2) 濃石炭酸

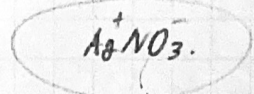
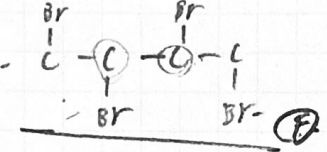
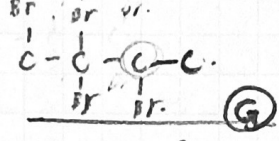
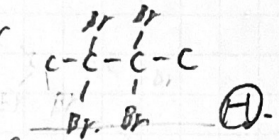
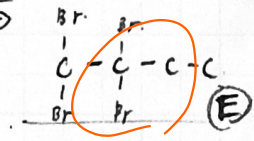
2-14

(a) A, D, C, D, E

C_4H_6

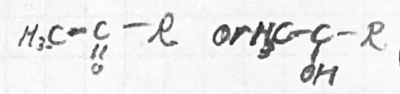


$+ 2 Br_2$

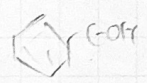


Aの試

水酸化



4-12 C_7H_8O



問1 操作1で発生した気体 H_2

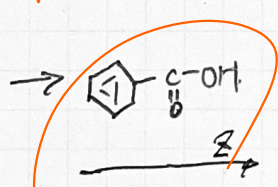
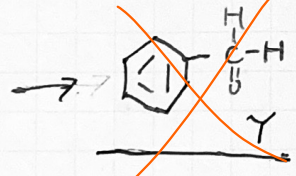
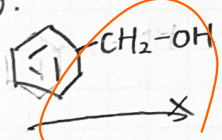
$\rightarrow$ 金属ナトリウム反応 $-OH$ or $-COOH$

問2 フェニル類の検出

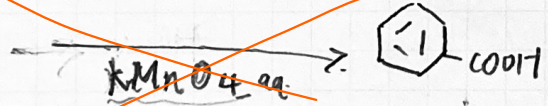
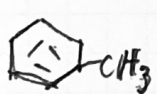
塩化鉄(III) $FeCl_3$

青色 ~ 青紫色

問3.



問4.



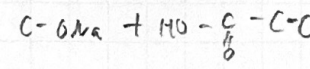
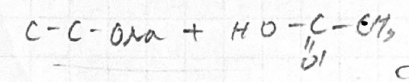
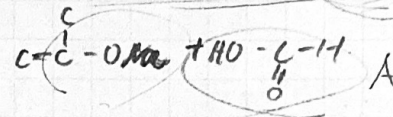
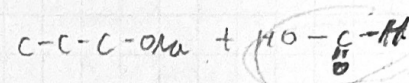
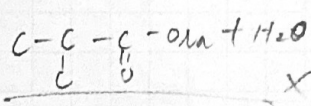
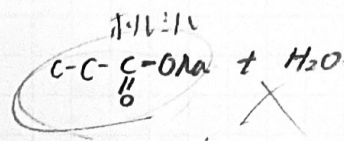
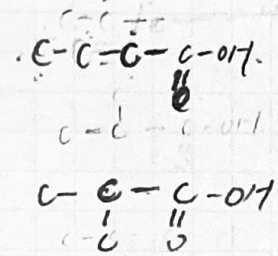
反応式

3-15
C₄H₈O₂

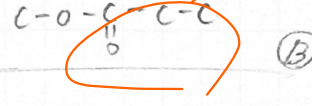
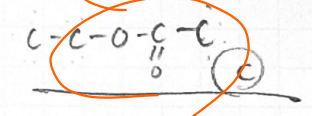
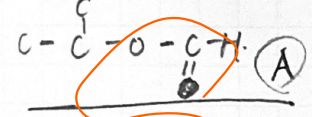
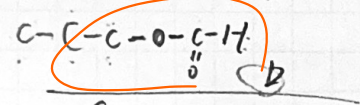
カルボン酸エステル

NaOH 中和

木炭 3-16
A B A C CH₃-C(=O)-H₂C(=O)-OH



72%



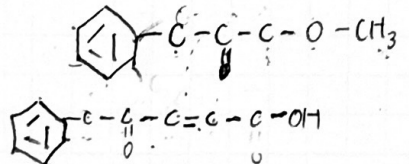
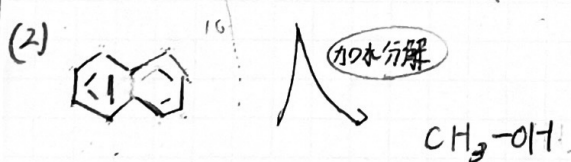
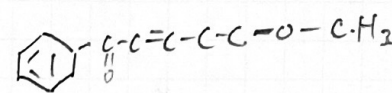
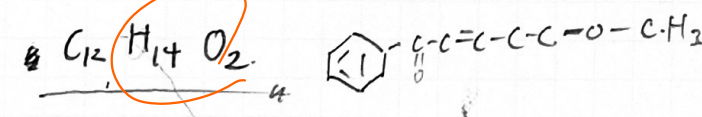
4-18

(1) C H O
75.8% 7.4% 16.8%

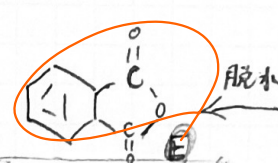
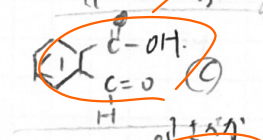
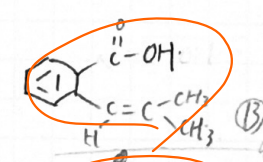
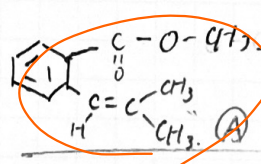
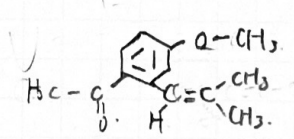
$\frac{75.8}{12} : \frac{7.4}{1} : \frac{16.8}{16}$

$6.15 : 7.4 : 1.05 \quad 19$
 $153 : 18 : 26 \quad 2 \times 7 + 16 = 95$

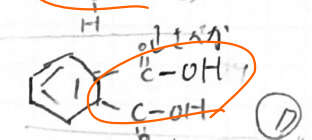
$14 : 15 : 2 \quad \frac{48}{168} \approx 200$
 2×5



$U = \frac{1 \times 2 + 2 \times 14}{2} = 6$
 $\sqrt{42} \approx 6.48$
 42



オゾン分解 → $\text{CH}_3-\text{C}(=\text{O})-\text{CH}_3$



2-②

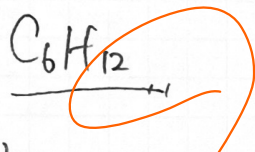
$$\frac{4}{160} = 0.025 \text{ mol} \quad \frac{3.21}{274} = 0.15 \text{ mol}$$

CH_2

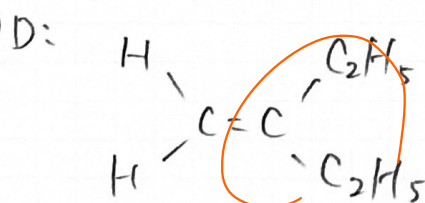
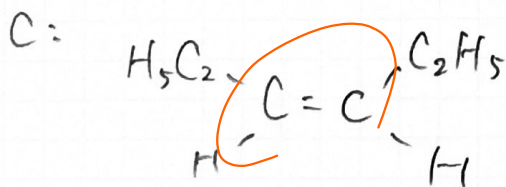
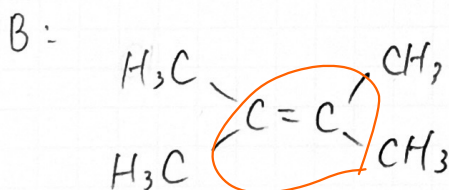
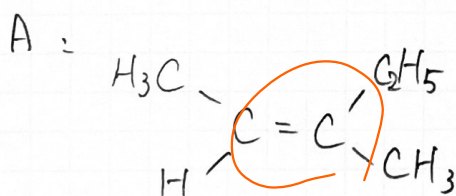
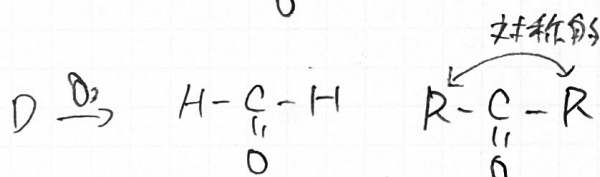
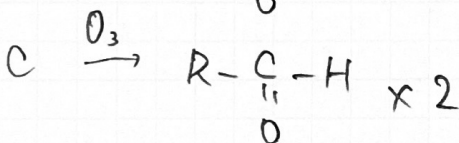
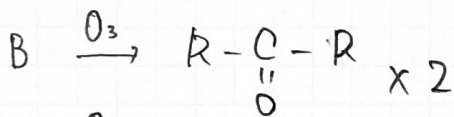
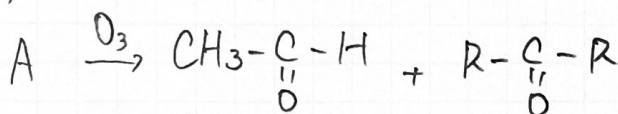
$$\frac{0.15}{n} \text{ mol of } (\text{CH}_2)_n \text{ is } 0.025 \text{ mol}$$

of Br_2 is added.

$$\Rightarrow \frac{0.15}{n} = 0.025 \quad \therefore n = 6$$

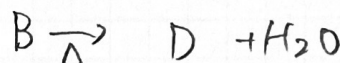
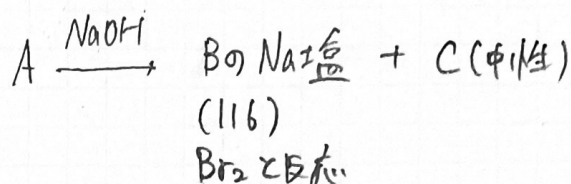


(2)



(3) A, C

3-⑦



C (アルコ-ILX) (還X) (EO)

$$(1) \text{C} : \text{H} : \text{O}$$

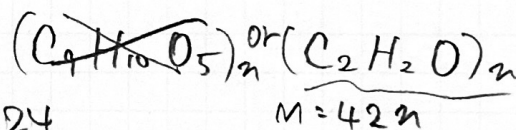
$$= \frac{53.8}{12} : \frac{5.1}{1} : \frac{41.1}{16}$$

$$= 4.5 : 5.1 : 2.5$$

$$= 9 : 10.2 : 5$$

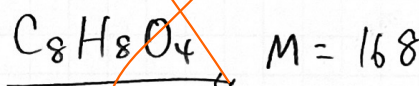
$$\frac{4.4}{12} \quad \frac{2.5}{16}$$

$$\frac{4.4}{12} \quad \frac{2.5}{16}$$



$$\frac{24}{16} \quad \frac{2}{42}$$

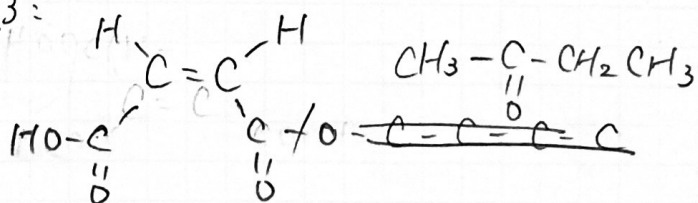
$$n = 4$$



(2)

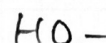
$$\textcircled{\times} = \frac{16 + 2 - 8}{2} = 5$$

B:

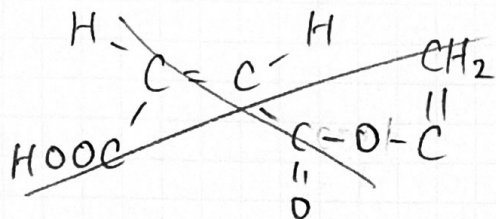


$$48 + 64 + 4$$

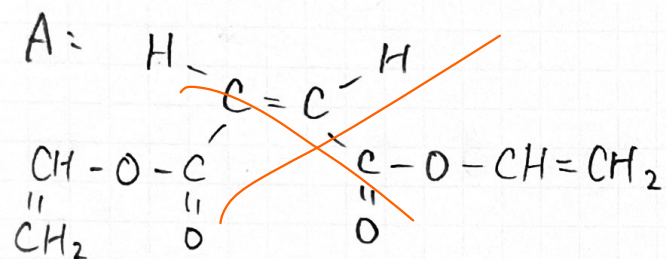
$$= 116$$



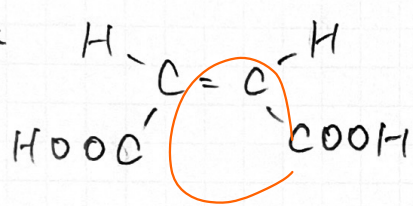
B:



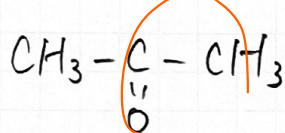
A:



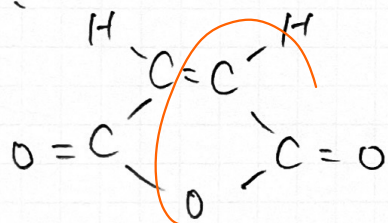
B:



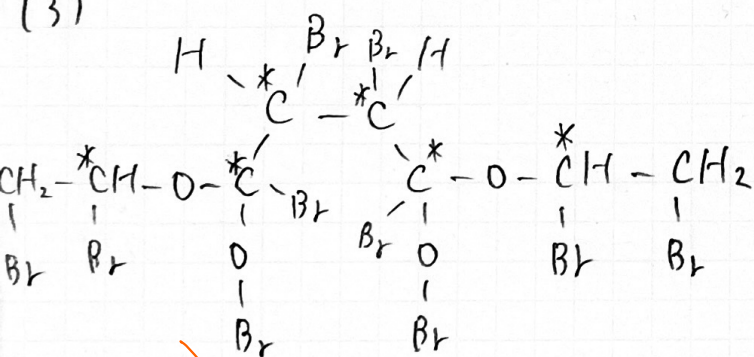
C:



D:



(3)



~~$$2^6 = 64$$~~

3-10

$$\text{C}_4\text{H}_{10}\text{O} \quad \textcircled{\neq} = \frac{8+2-10}{2} = 0$$

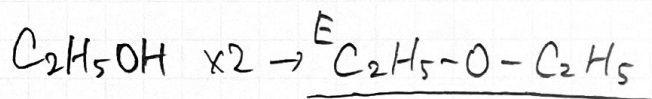
A (アミル-10)

B (")

C (")

D (") $\xrightarrow[\Delta]{\text{H}_2\text{SO}_4}$ F (2重結合あり)

E (アミル-10X)



(1)

A $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$

B $\text{CH}_3\text{CH}_2\text{CH}(\text{OH})\text{CH}_3$

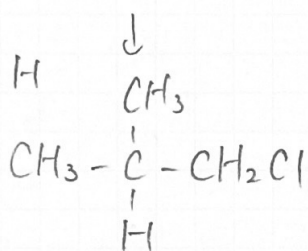
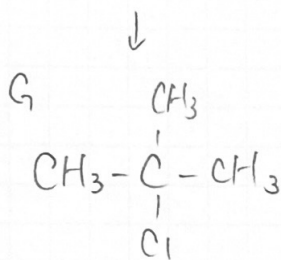
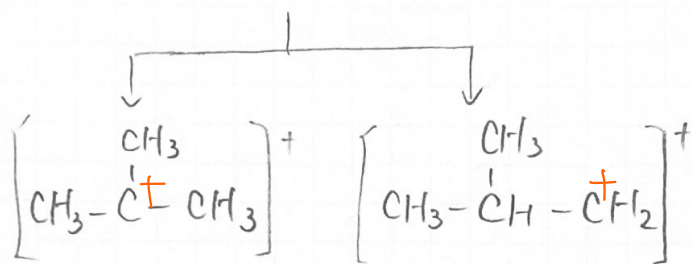
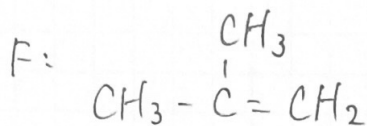
D $\text{CH}_3\text{C}(\text{OH})(\text{CH}_3)_2$

C $\text{CH}_3 - \text{O} - \text{CH}_2\text{CH}_2\text{CH}_3$

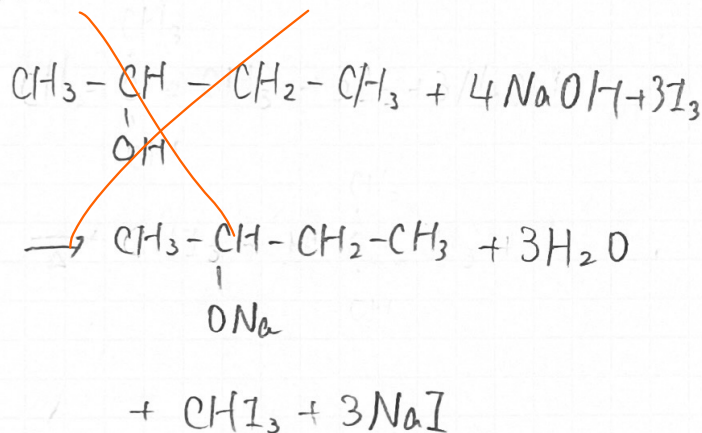
E $\text{C}_2\text{H}_5 - \text{O} - \text{C}_2\text{H}_5$

3-10

(2)



(3)

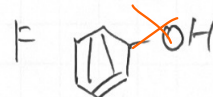
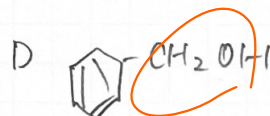
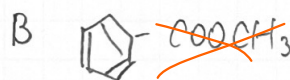
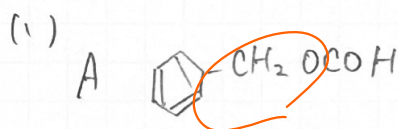
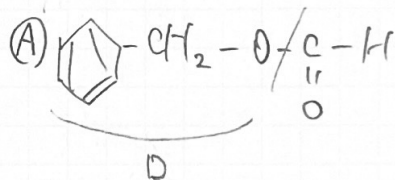
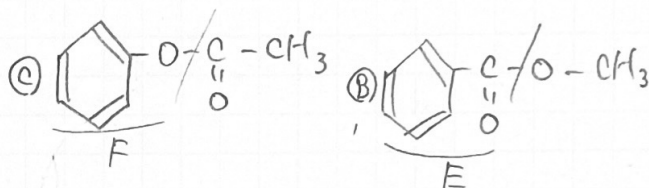
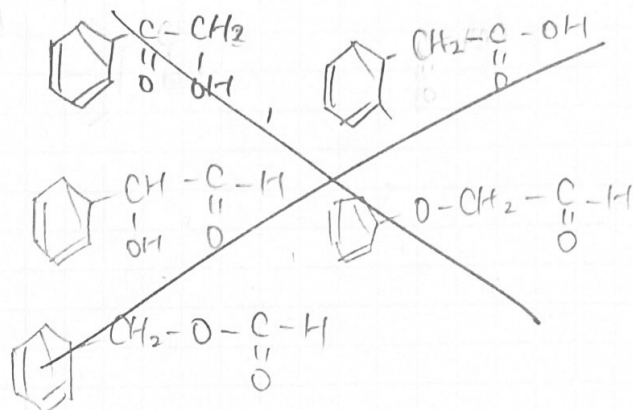


(4) B

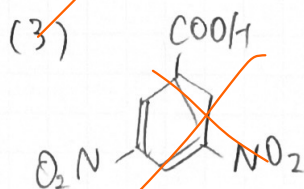
(5) I-716 88.6

4-17

$\text{C}_8\text{H}_8\text{O}_2$ $\text{DB} = \frac{16+2-8}{2} = 5$

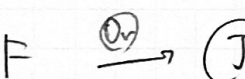
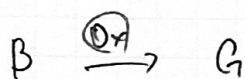
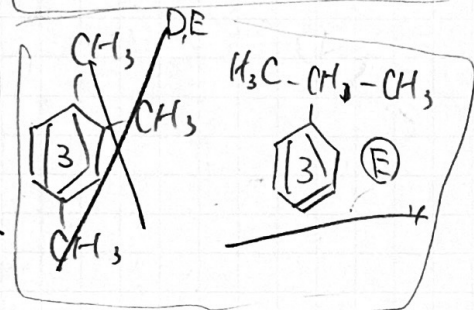
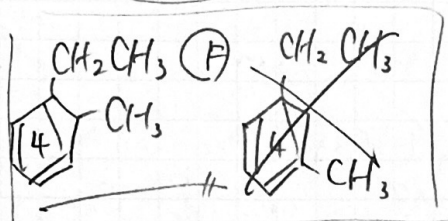
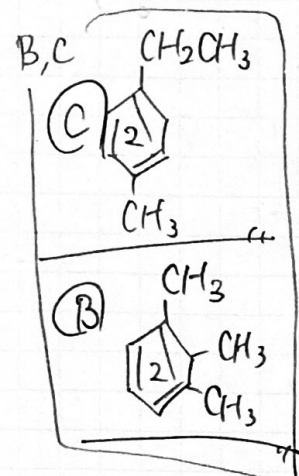
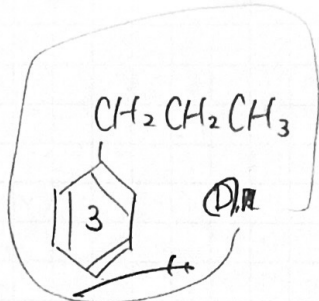
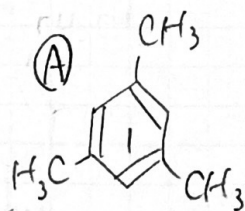


(2) F



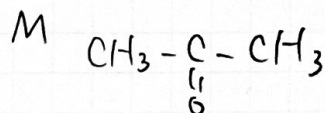
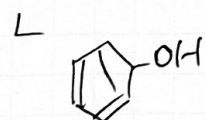
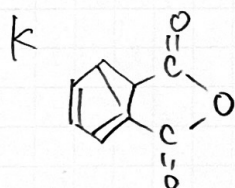
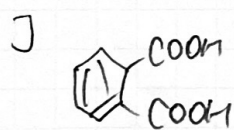
3,5-ジニトロ安息香酸

4-10



異置換体

同置換体



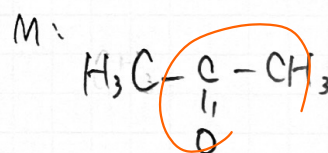
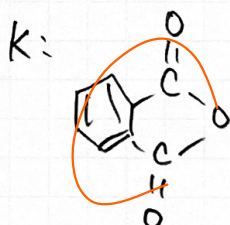
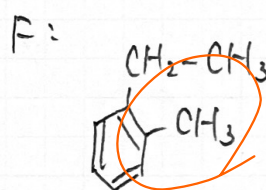
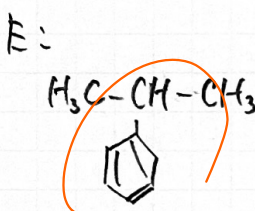
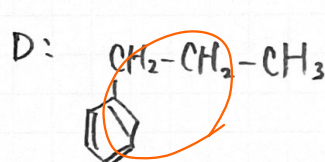
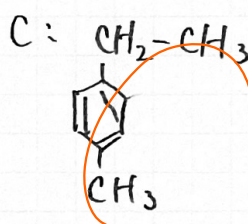
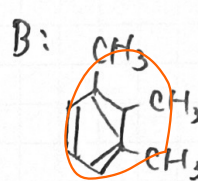
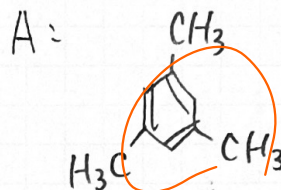
(1)

(A): 1,2-エタンジオール

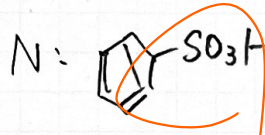
(K): 酸素

(A): 酸素

(2)



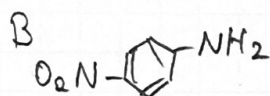
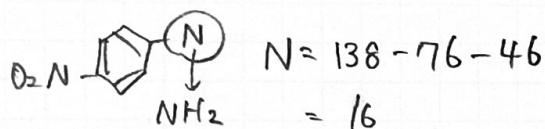
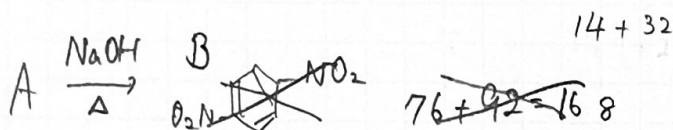
(3)



4-20

A: (カルボン酸)

$$\textcircled{A} = \frac{22 + 2 - 12 + 2}{2} = 7$$



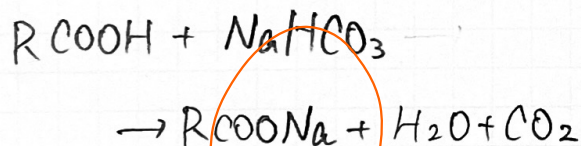
(1) C : H : N : O $\begin{array}{r} 132 \\ 12 \\ 28 \\ + 80 \\ \hline 252 \end{array}$
 $= 132 : 12 : 28 : 80$

C ... 52.4% O ... 31.7%

H ... 4.8%

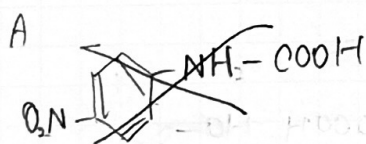
N ... 11.1%

(2)

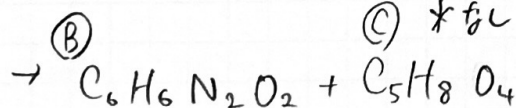
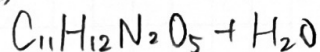


(3)

$17 = 4 + 1 + 2$
 炭 水素 酸素



(A) * 17

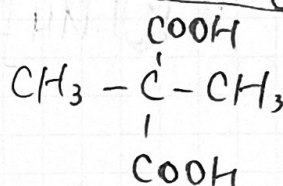
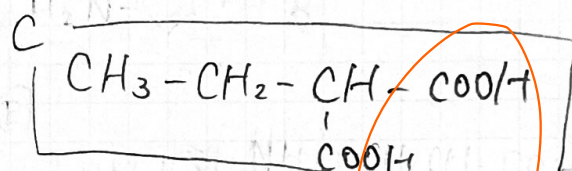
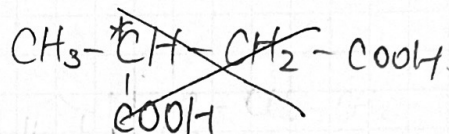
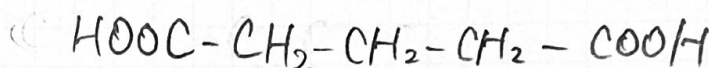
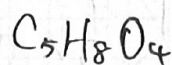


* 16

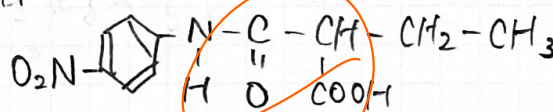
(C) * 16

-COOH 2 x 2

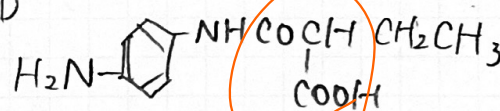
$$\textcircled{C} = \frac{10 + 2 - 8}{2} = 2$$



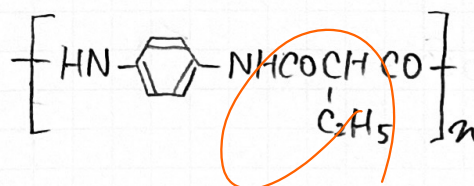
A



D



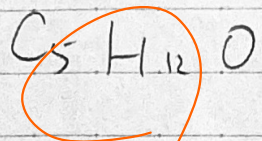
E



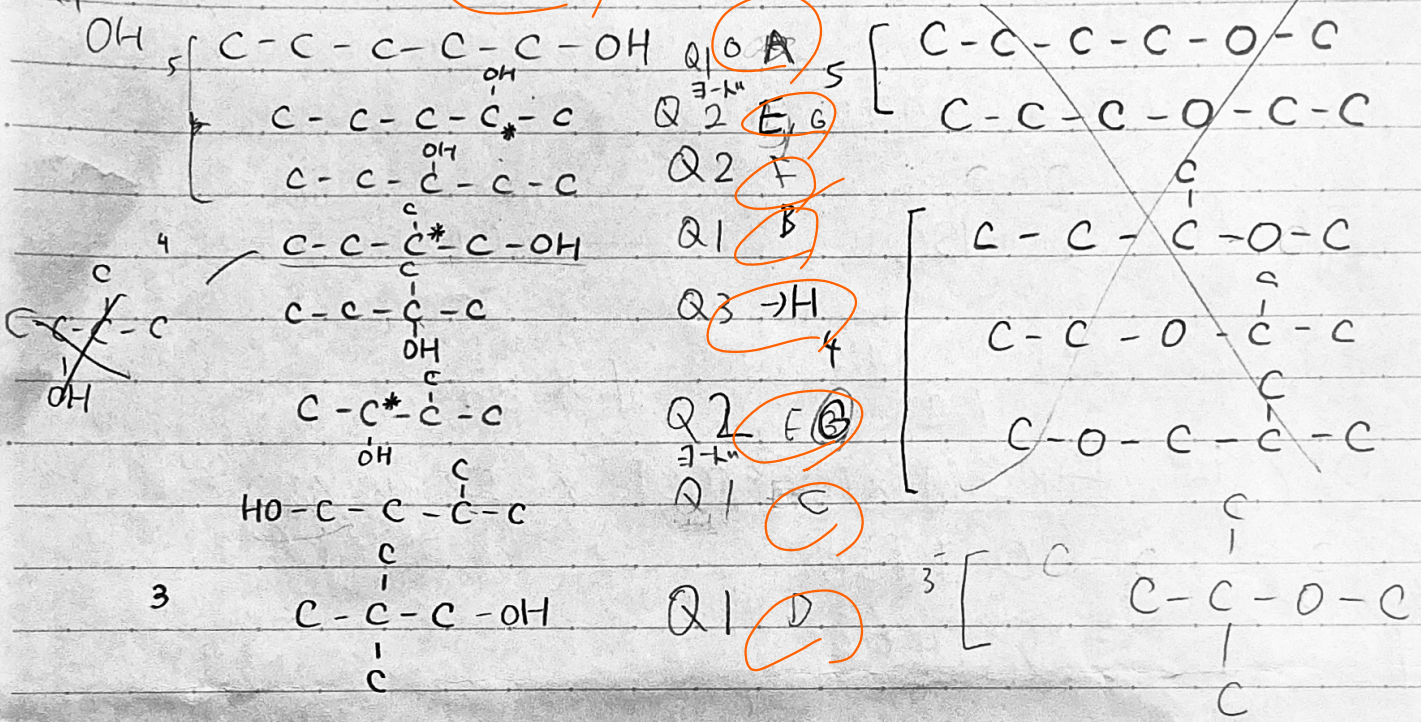
(4)

アミド結合が塩酸により加水分解
 してほろろ。

3-2

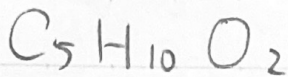


1)



(2) 6

3-15



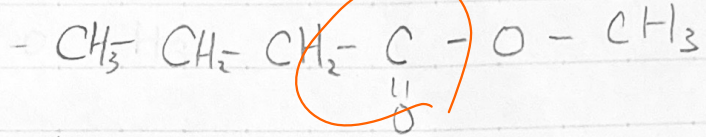
分子量
60+10+32

$$\begin{array}{r} 96 \\ \times 13 \\ \hline 288 \\ 96 \\ \hline 1248 \end{array}$$

$$\begin{array}{r} 0.962 \\ 106 \overline{) 1020} \\ \underline{959} \\ 660 \\ \underline{636} \\ 240 \end{array}$$

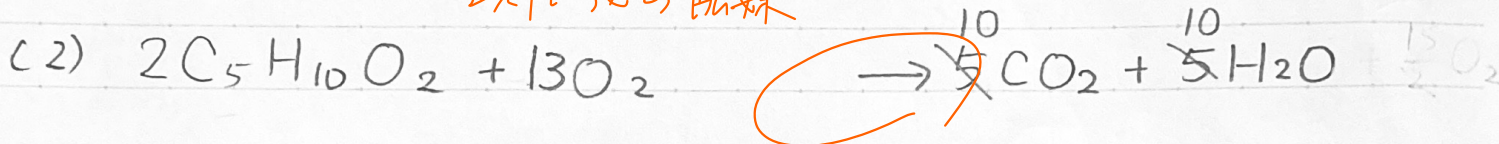
反応式

(1) 酢酸 $\times$ 4L



硫酸: 脱水の促進のため.

エステル化の触媒



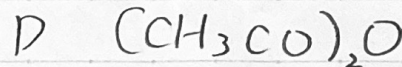
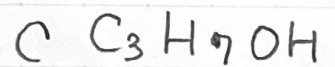
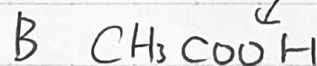
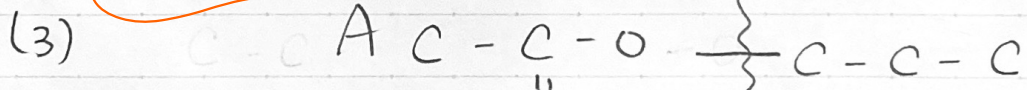
酢酸 $\times$ 4L $\rightarrow \frac{1.02 \times 10^{-1}}{102} = 1 \times 10^{-3} \text{ mol}$

$O_2 (L)^{12} \quad \frac{13}{2} \times 0.1 \times 10^{-3} \text{ mol} \times 22.4 \text{ L/mol}$

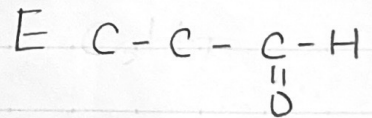
$= 0.1456 \text{ L}$

$\approx 0.146 \text{ L}$

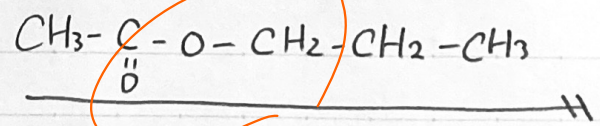
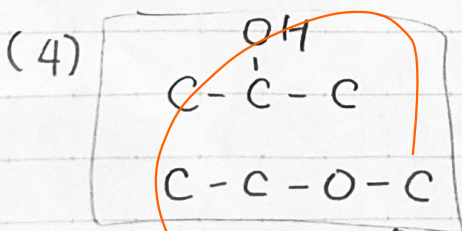
$$\begin{array}{r} 224 \\ \times 13 \\ \hline 672 \\ 224 \\ \hline 2912 \end{array}$$



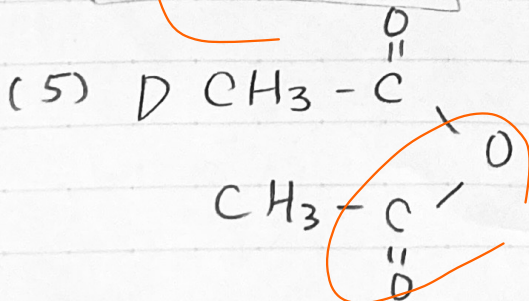
$\downarrow$ 酸化



(3) B 酢酸 C1-7'口101-L A



(6) 水酸基

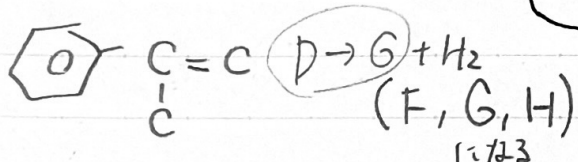
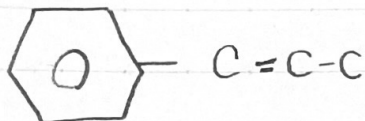
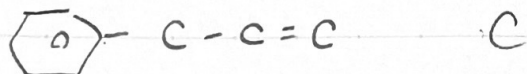


$$C_3H_4$$

②

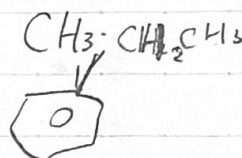
C_3H_6

(1) C_3H_5 ($EC \div$)



教方商の如く
異性体と云ふ。

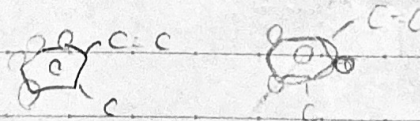
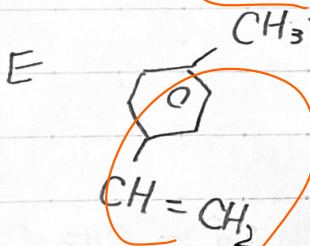
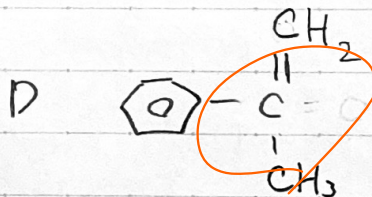
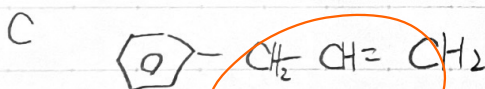
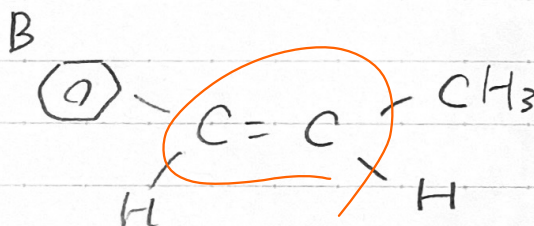
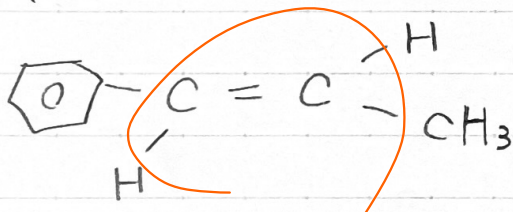
P
E
J
H



Chc ch-on

(1) ~~6種類~~

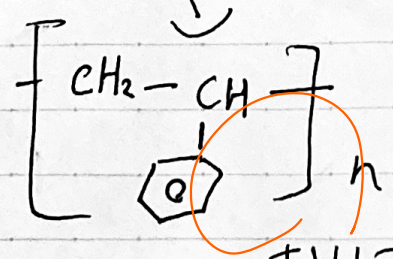
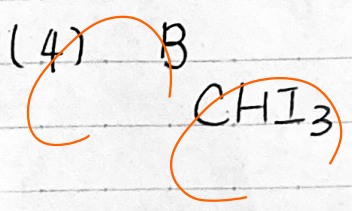
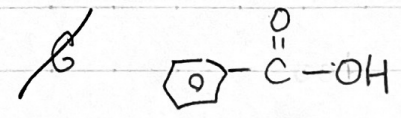
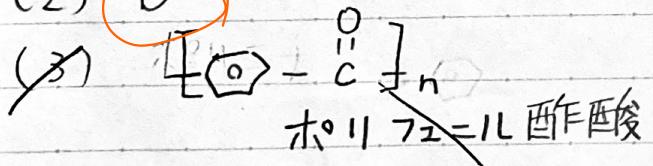
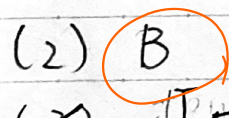
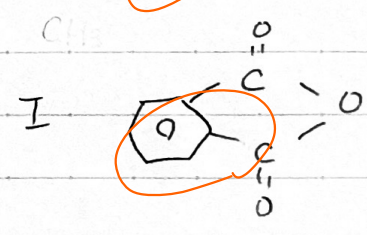
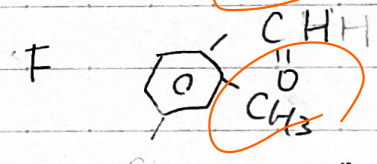
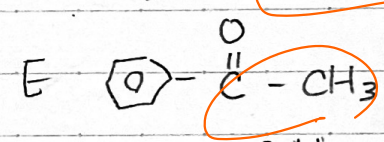
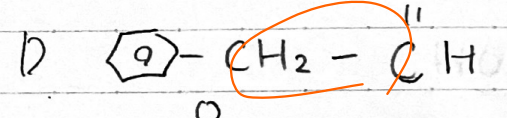
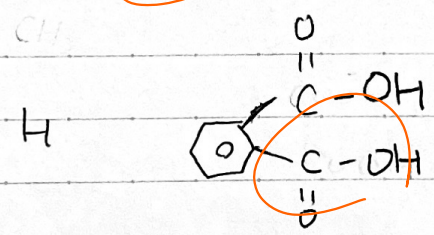
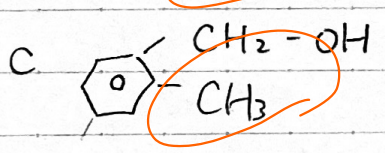
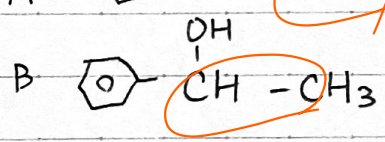
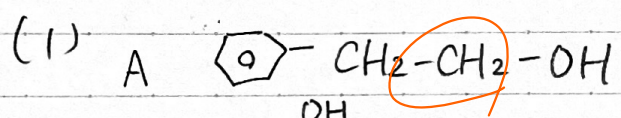
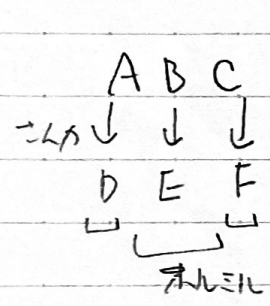
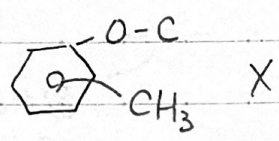
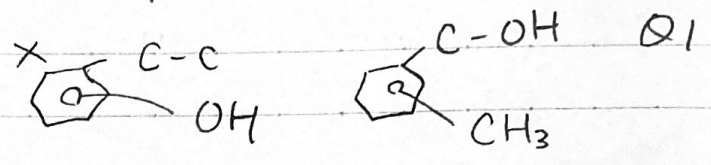
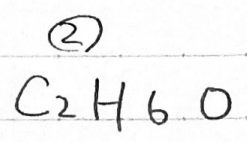
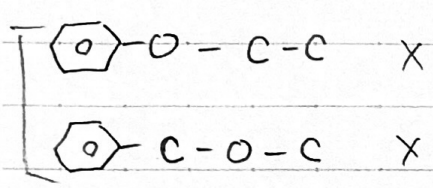
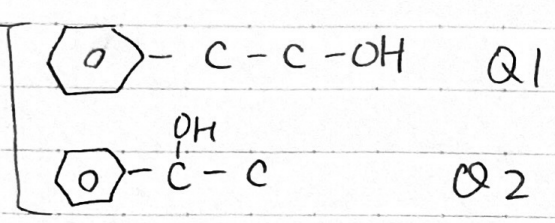
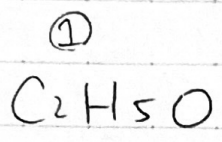
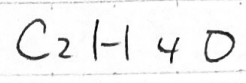
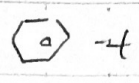
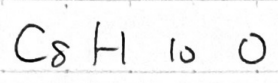
(2) A



(3) $6 + 8 \cdot 2 = 16$

$$6 + 10 + 6 = \underline{22}$$

4-15

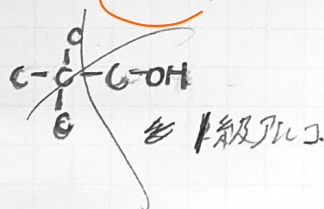
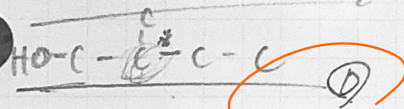
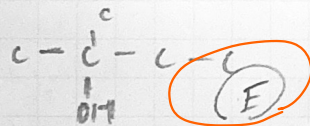
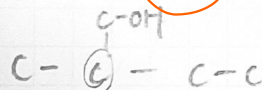
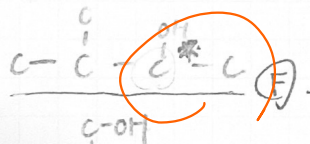
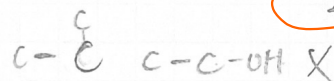
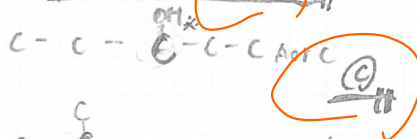
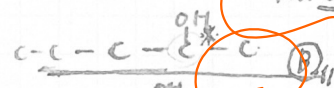
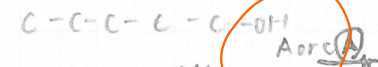


ポリスチレン

脱水 → 濃硫酸 + 加熱
 $KMnO_4$ + 加熱
側鎖がカルボン酸に

まさか
理由

3-12 $C_5H_{12}O$ アルコール



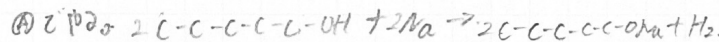
ABC 枝分かれし。

DF ヨード試液

E のみ 脱離しにくい

A, D の酸化はのち $AgNO_3$ が反応。
ホルミル基。

問2.



$n_2 \frac{0.3}{23} = 0.01300706$

$\frac{10}{16+12+60} = 0.125$

$0.013 \dots$

$\frac{0.3}{23} \times \frac{1}{2} \times 22.4 = 0.736$

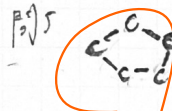
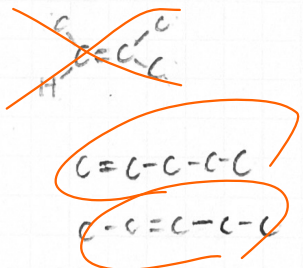
0.14608

$0.15 L$

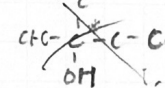
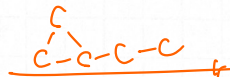
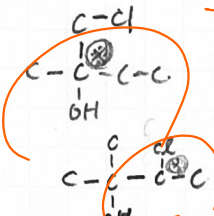
問3. 銀鏡反応



問4. C_5H_{10}

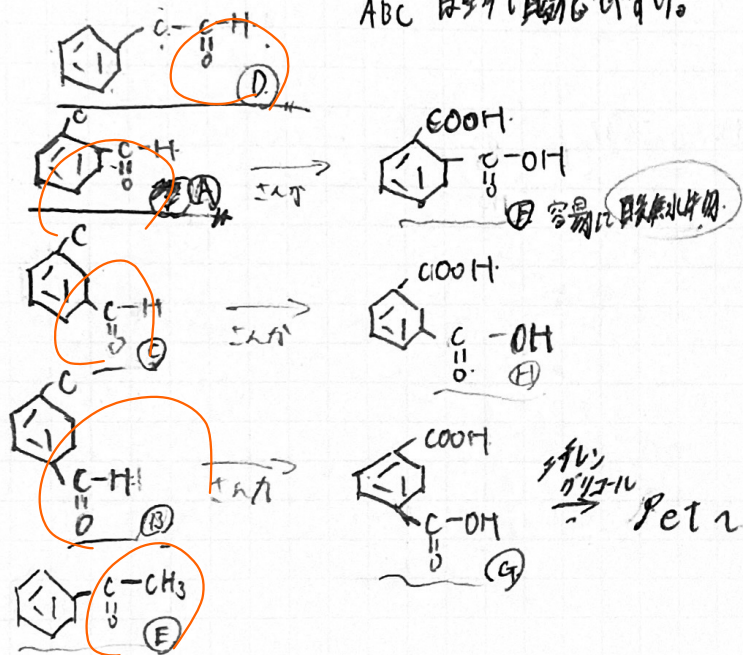


問6.



4-5. C_8H_8O カルボニル基 $-C(=O)-$

ABC はそれぞれ酸化しやすい。



F, H は官能基の配置から分子内

水素結合が起るから G はハロの配置であり
他の分子と水素結合を起しにくいから

G は F, H より

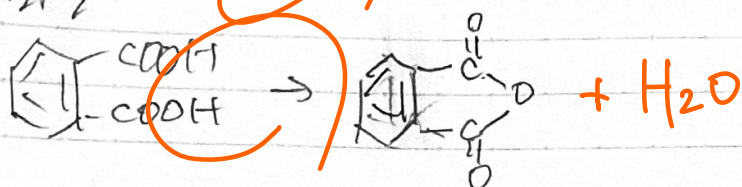
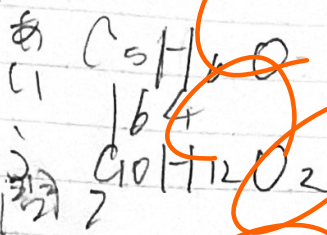
対称性が低いから、

結晶化しやすい

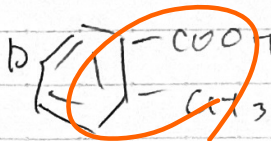
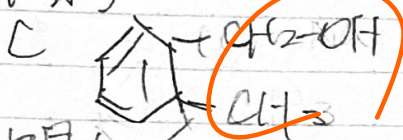
→ 融点高い

4-19

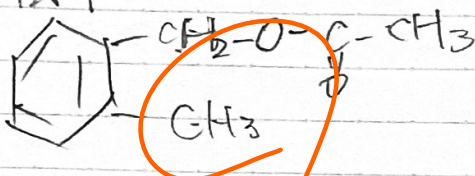
問1



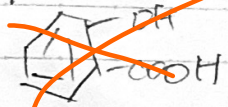
問3



問4

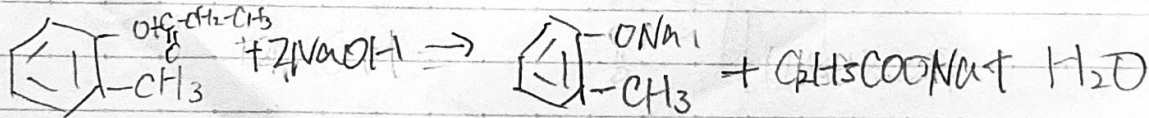


問5

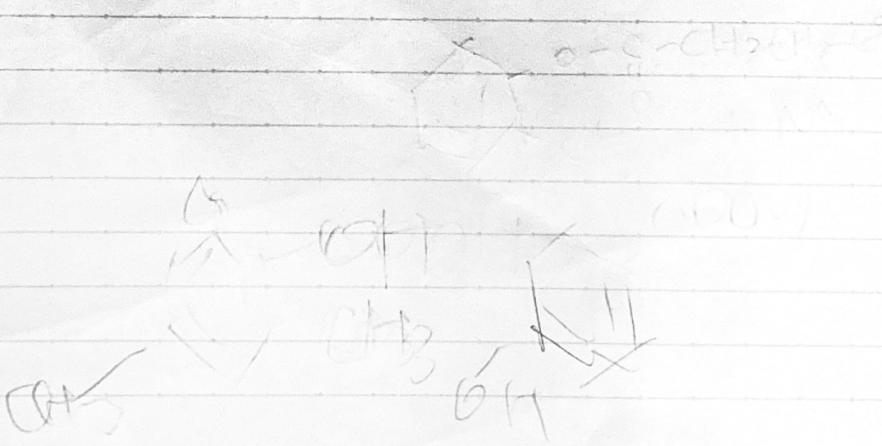


105位のやつです

問6

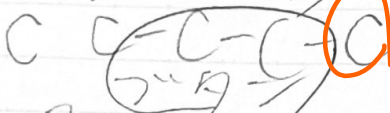
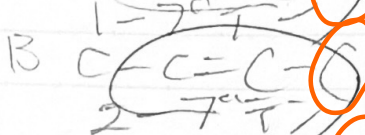
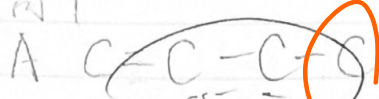


105位のやつ



2-3

問1

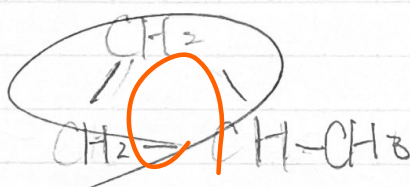
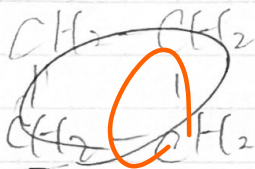


問2

D 2種類

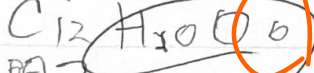
E 1種類 3種類

問3



3-12

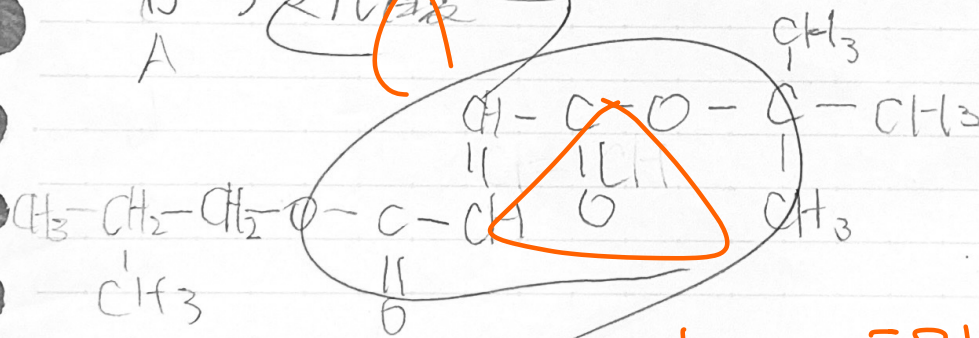
問1



問2



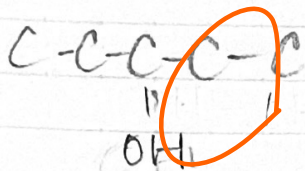
A



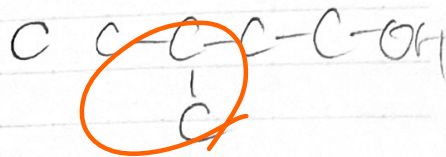
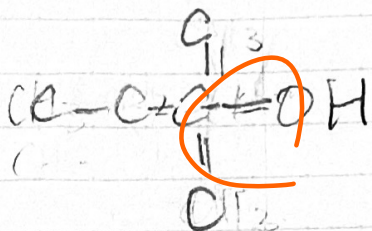
シス・トランス区別

答えはトランス型

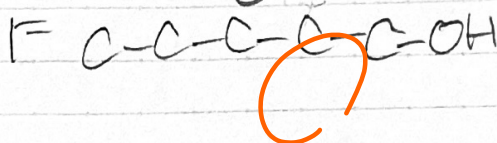
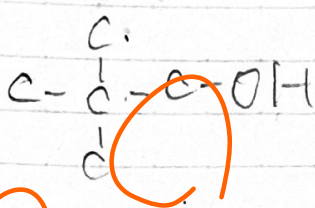
3-~~III~~
(1) A



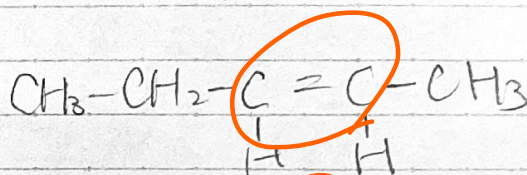
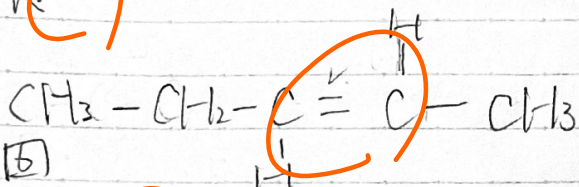
B



D

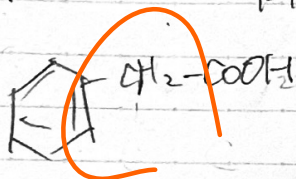


(2) K
(3)

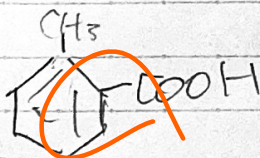


4-~~II~~

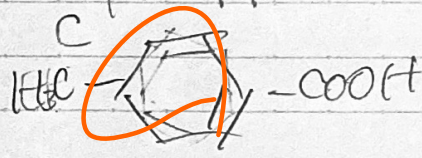
A



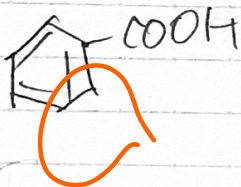
B



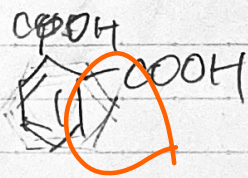
C



P



E



I

